

Cadangan Panas Bumi

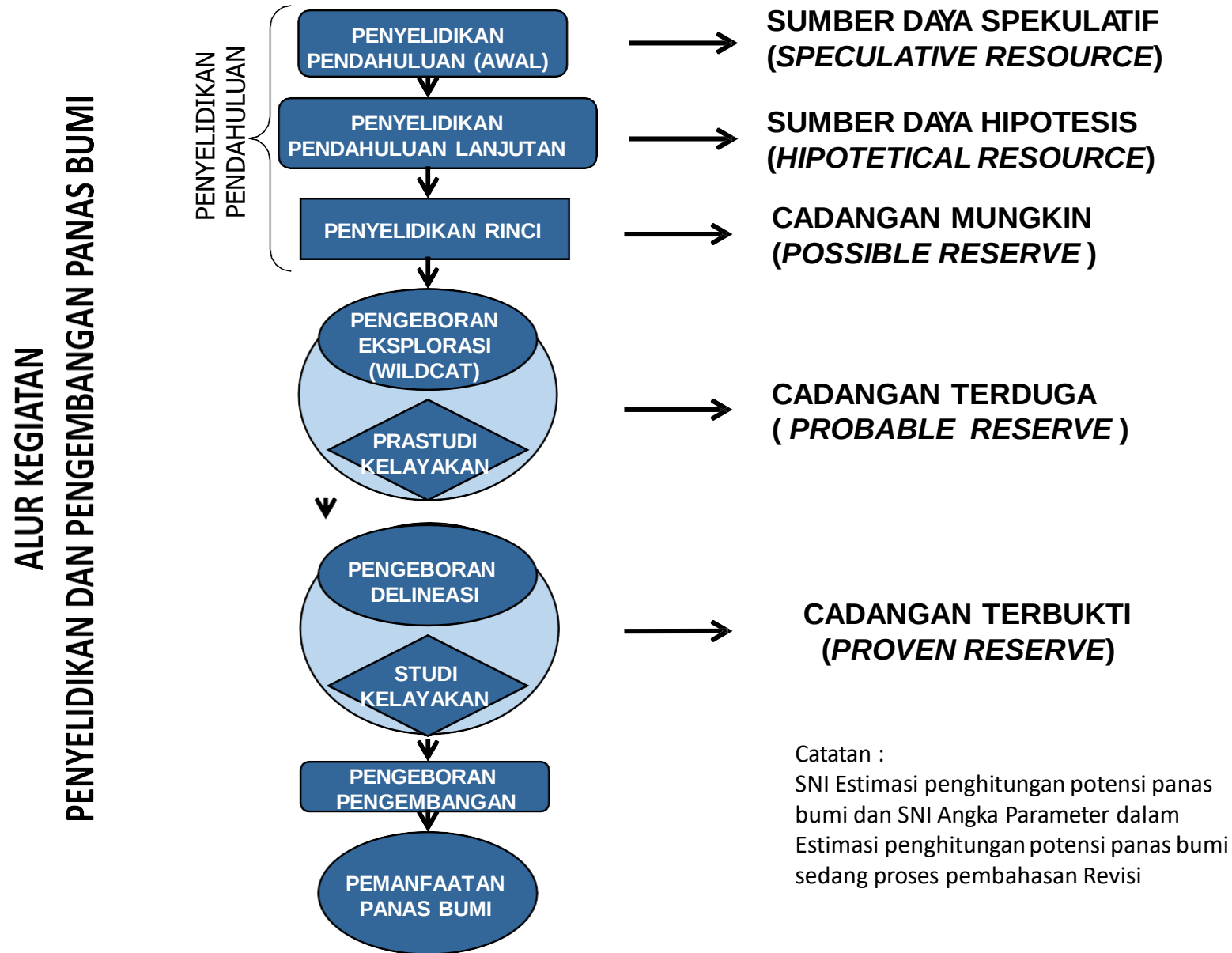
Ali Ashat
Kamis, 29 November 2024



Metoda Perhitungan

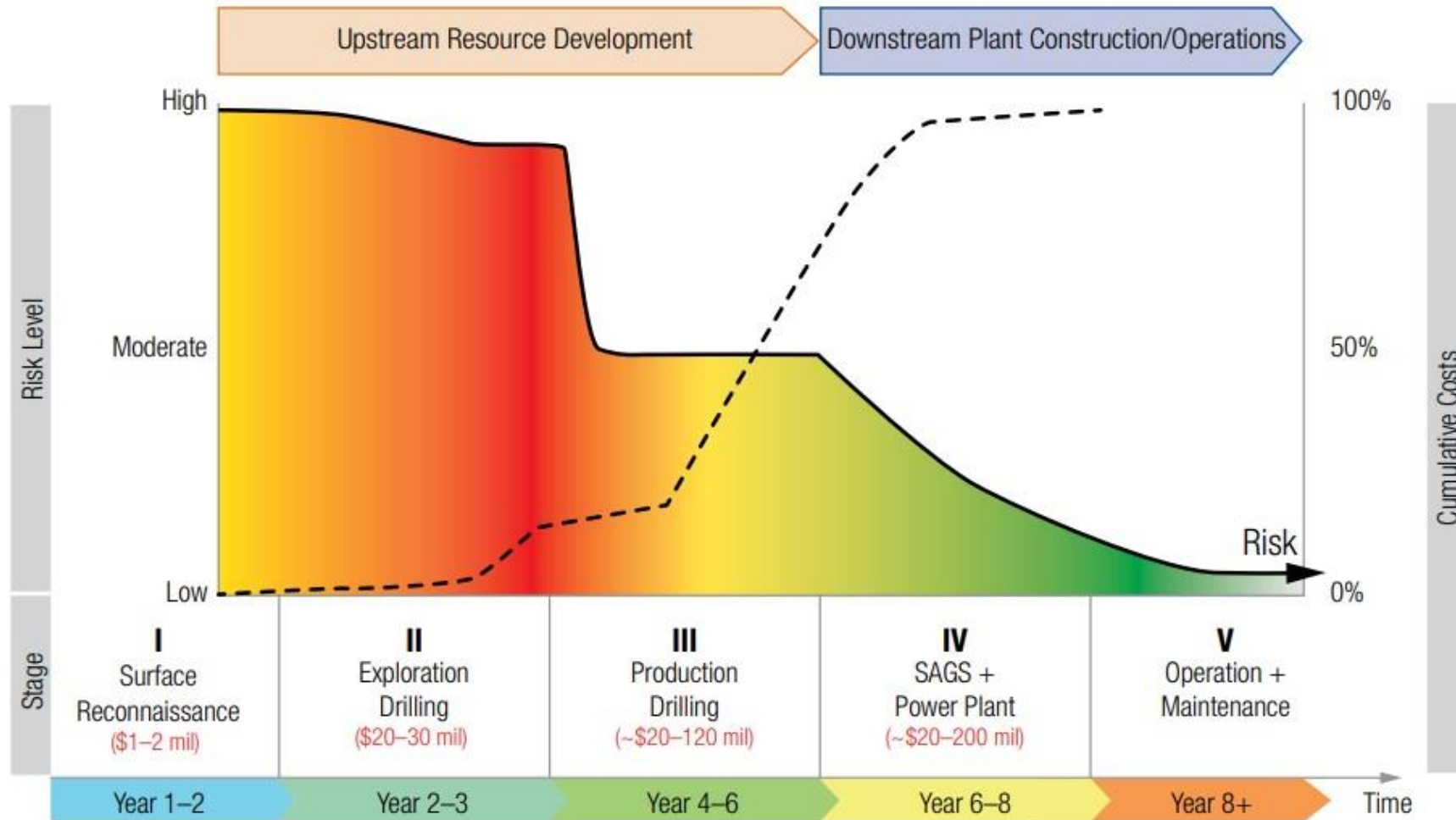
1. Heat Flux Method
2. Metoda Perbandingan (Areal Method)
3. Metoda Volumetrik (Heat Stored Method)
4. Decline Curve Analysis
5. Metoda P/Z
 - a) Dry Steam
 - b) Metoda Falling Liquid Level
 - c) Metoda Constant Liquid Level
6. Metoda Simulasi Reservoir

TAHAPAN EKSPLORASI DAN PENGEMBANGAN GEOTHERMAL



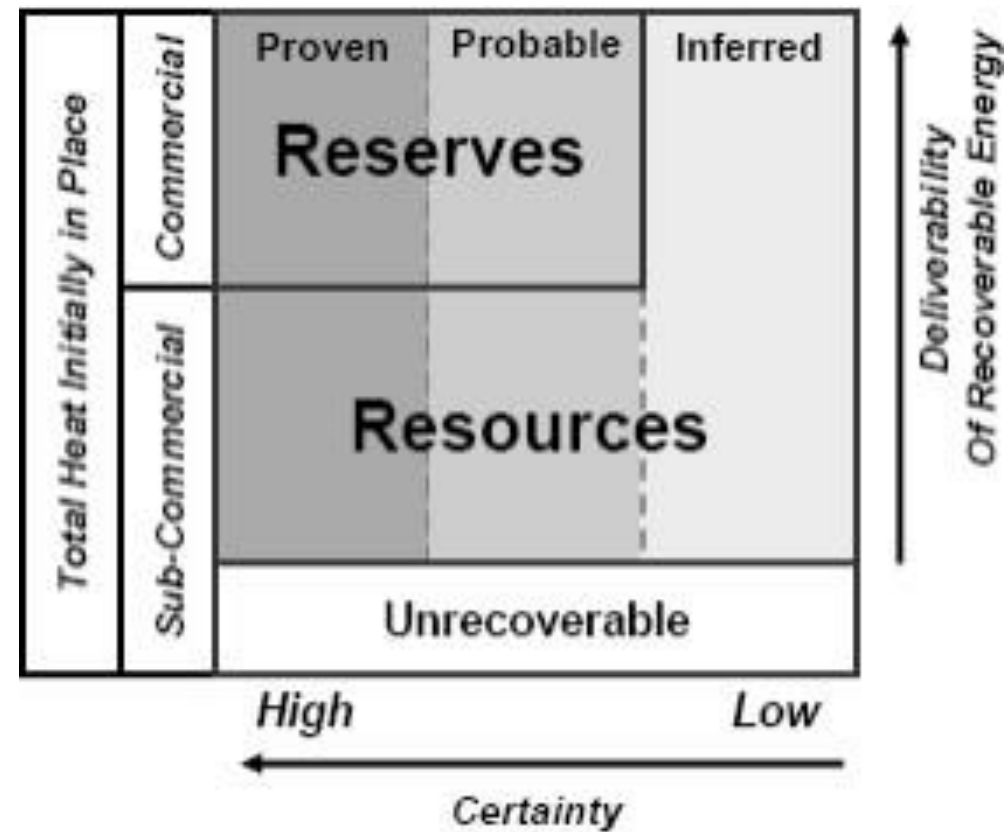
KLASIFIKASI SUMBER DAYA DAN CADANGAN
ENERGI PANAS BUMI INDONESIA
SNI 6009 : 2017

Figure 1 | A Conceptual Representation of Risks and Costs during the Different Stages of a Geothermal Development



Source: Adapted from Geothermal Handbook (ESMAP 2012).

Klasifikasi Potensi Geotermal [Clothworthy, 2006]



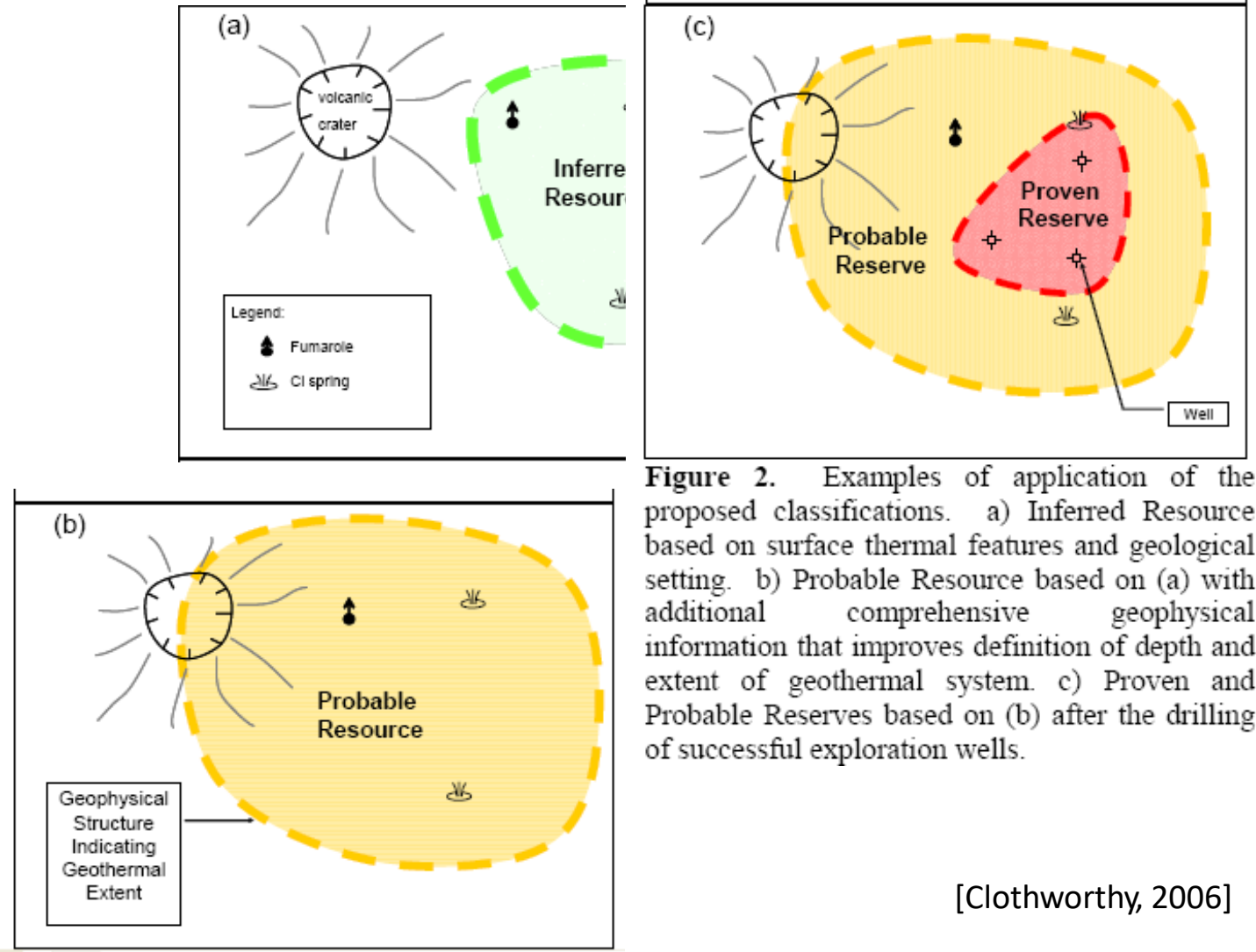
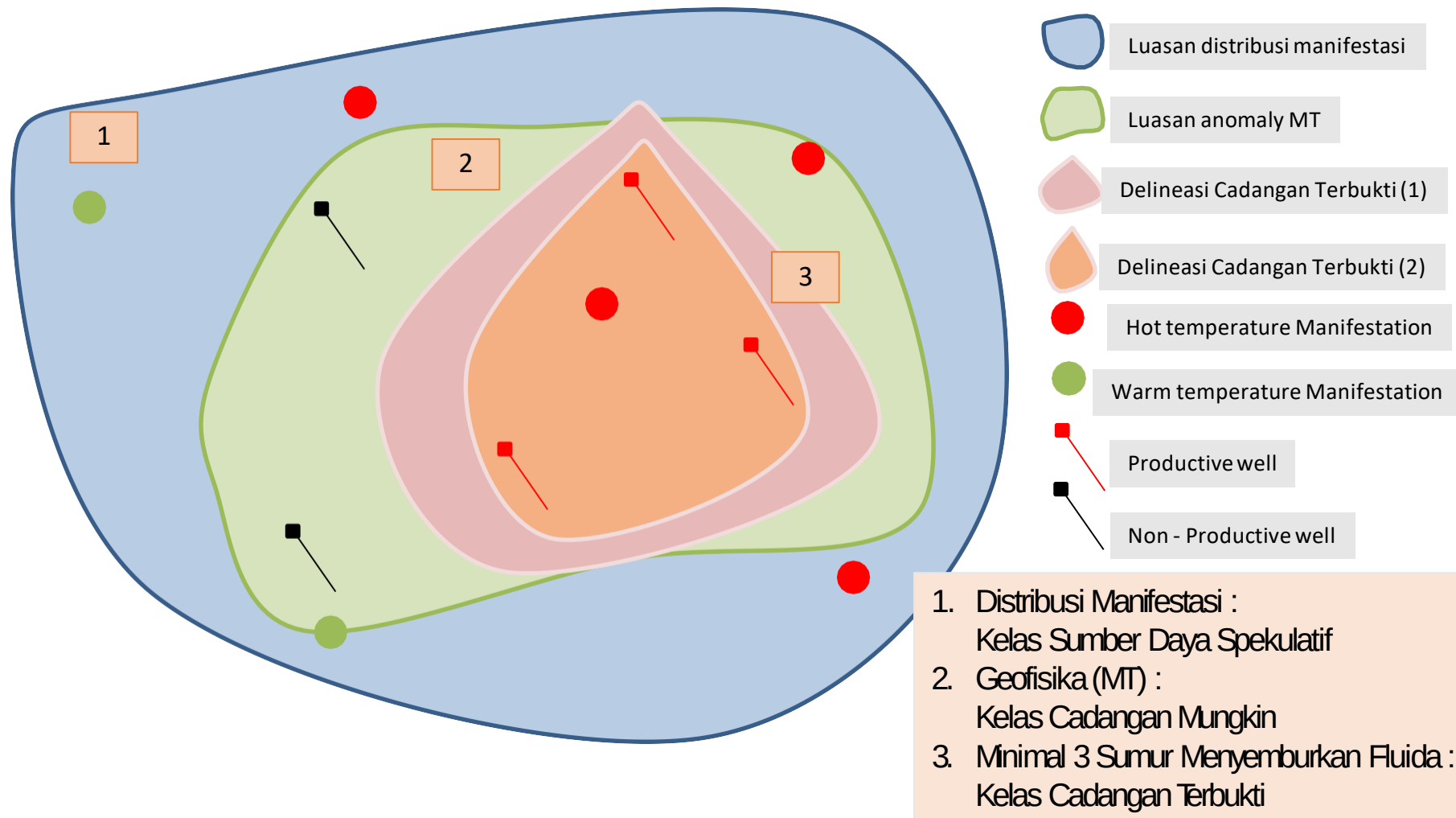


Figure 2. Examples of application of the proposed classifications. a) Inferred Resource based on surface thermal features and geological setting. b) Probable Resource based on (a) with additional comprehensive geophysical information that improves definition of depth and extent of geothermal system. c) Proven and Probable Reserves based on (b) after the drilling of successful exploration wells.

[Clothworthy, 2006]



HEAT FLUX METHOD (ESTIMASI BERDASARKAN LAJU ALIRAN PANAS)

- Dari hasil pengukuran temperatur tanah dapat diperkirakan besarnya aliran panas yang terjadi secara konduksi (Q_e):

$$Q_e = -K \left(\frac{dT}{dz} \right) A$$

K = Konduktivitas panas batuan ($W/m^{\circ}K$)

A = Luas daerah (m^2)

(dT/dz) = Gradien temperatur ($^{\circ}C/m$)

- Dari harga laju aliran massa dan temperatur air dapat dihitung besarnya aliran panas ke permukaan yang terjadi secara konveksi,

$$Q_e = m_L h_L + m_v h_v$$

Q_e = Laju alir panas total (dalam kJ/detik atau kW)

h_L = Enthalpy air (kJ/kg)

h_v = Enthalpy uap (kJ/kg)

m_L = Laju alir massa air (kg/detik)

m_v = Laju alir massa uap (kg/detik)

KONDUKTIVAS PANAS BEBERAPA JENIS BATUAN

JENIS BATUAN	KONDUKTIVITAS (W/m.K)
Limestone	2.2 – 2.8
Slate	2.4
Sandstone	3.2
Bituminous coal	0.26
Rock salt	5.5
Gneiss	2.7
Granite	2.6
Gabbro	2.1
Peridotite	3.8

CONTOH: PERKIRAAN PANAS YANG HILANG MELALUI MANIFESTASI PANAS BUMI PERMUKAAN DI LAPANGAN KAMOJANG

JENIS MANIFESTASI	PANAS (MWt)
Semua fumarole	20.6
Kawah Pangasahan	4.2
Sumur nomor 3	8.2
Semua mata air panas/hangat	17.2
Mata air Cipangasahan	10.5
Mata air panas yang terletak 2.5 km di sebelah selatan Kawah Kamojang	0.2
Penguapan dari telaga dan kolam-kolam air panas/hangat	44.0
Telaga Kawah Manuk	22.1
Tanah Beruap	2.9
Lain-lain	25.2
TOTAL	97.4



Wairakei
New Zealand



Geyser USA,

$$H_{el} = A \times Q_{el}$$

H_{el} = Besarnya sumber daya (MW_e)

A = Luas daerah prospek panas bumi (km^2)

Q_{el} = Daya listrik yang dapat dibangkitkan
persatuan luas (MWe/km^2)



“Densitas Produksi”
Atau “Extraction Rate”

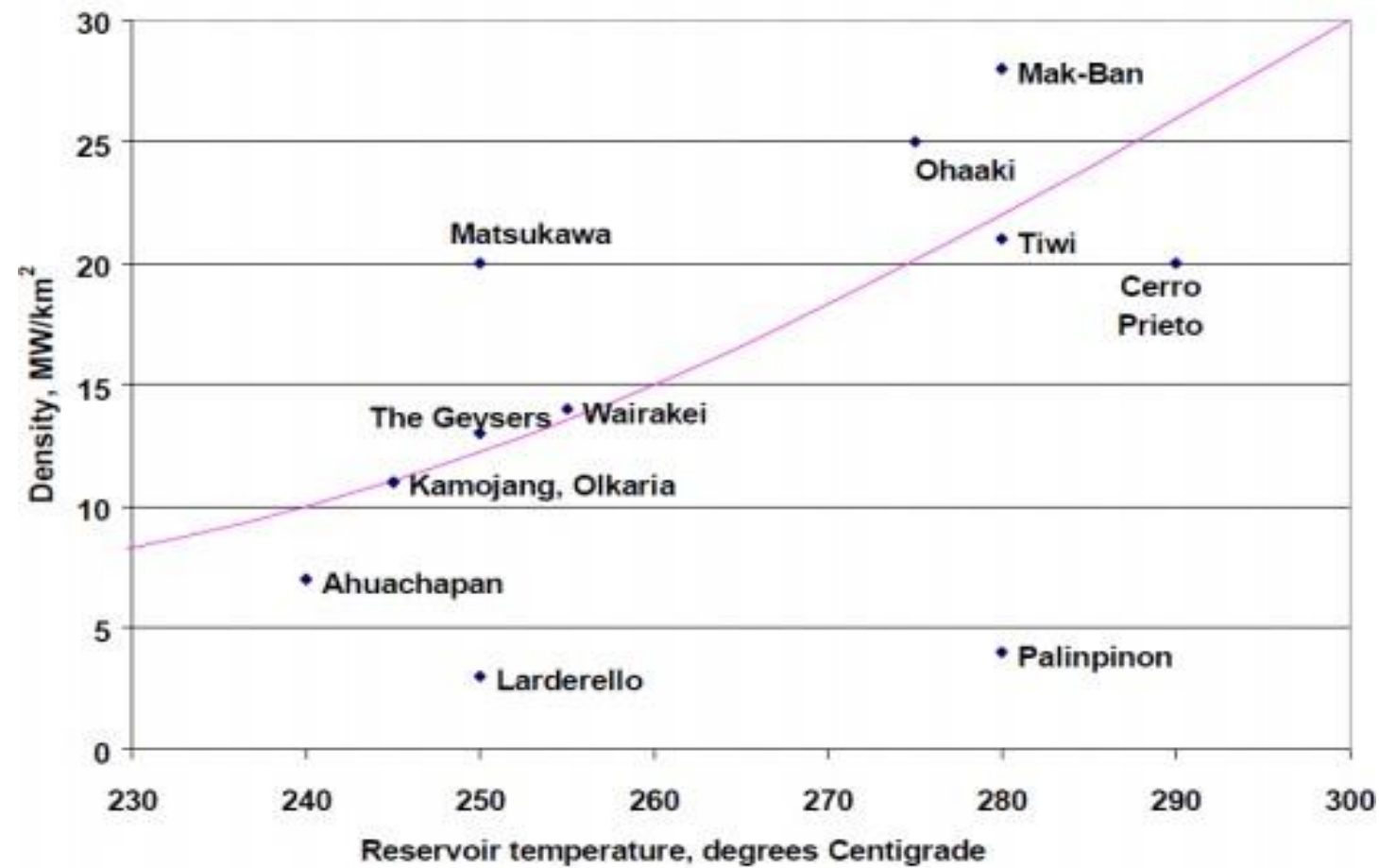
Tabel 1 – Nilai parameter rapat daya rata-rata pada metode perbandingan

Parameter	Temperatur rendah ($<125\text{ }^{\circ}\text{C}$)	Temperatur sedang ($125 - 225\text{ }^{\circ}\text{C}$)	Temperatur tinggi ($>225\text{ }^{\circ}\text{C}$)
Rapat Daya (MWe/km^2)	4 – 5	6 – 7	13 – 15

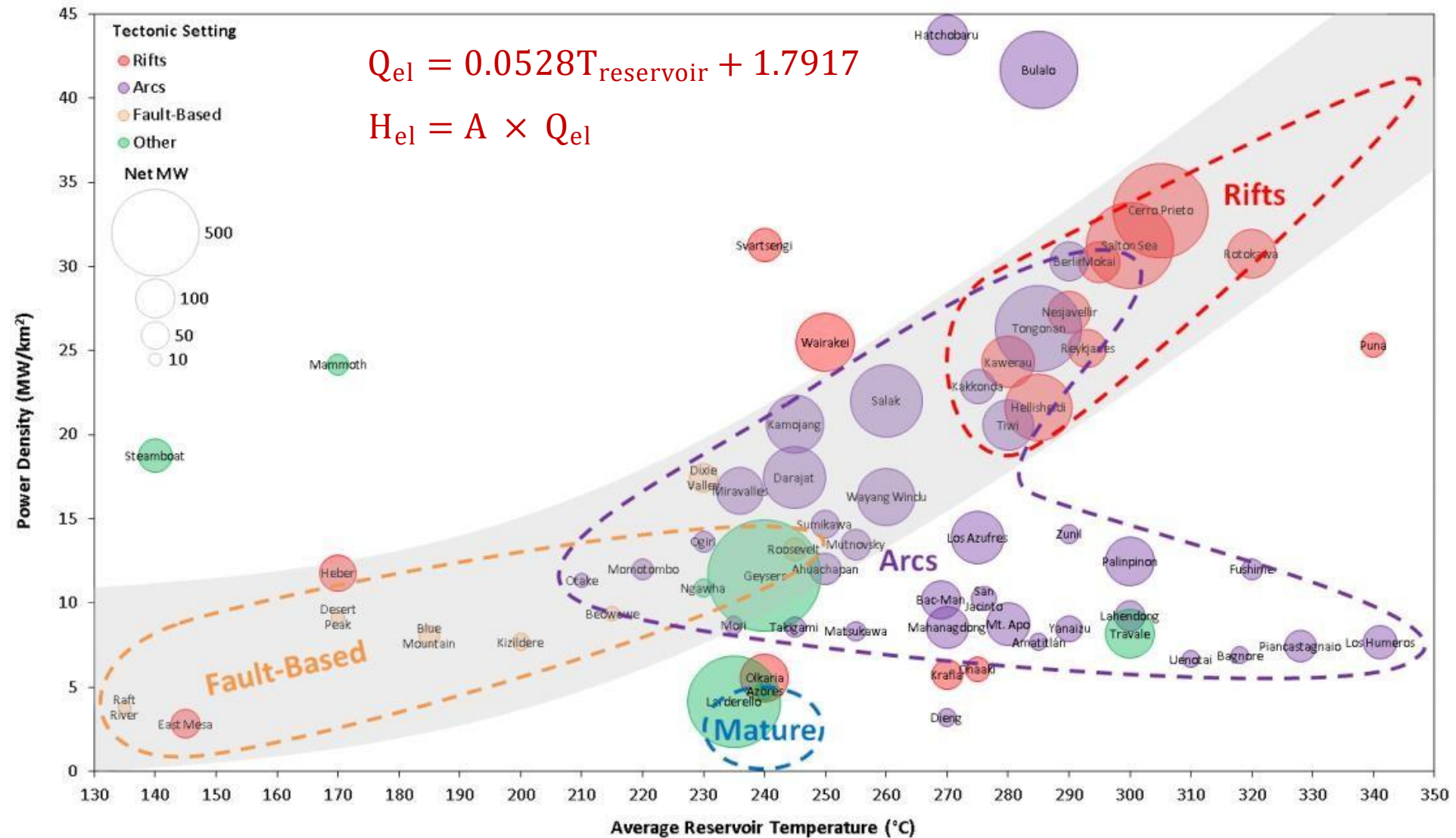
SNI 6482:2018 Parameter dalam estimasi potensi energi panas bumi

Lapangan	Kapasitas Terpasang (MWe)	Luas Area (km ²)	Daya Listrik/km ² (MW/km ²)
Bulalo	330	20	16.5
Tiwi	330	12	27.5
Tongonan	112.5	40	2.8
Palinpinon	112.5	47	2.4
Ahuchapan	125	6.4	19.5

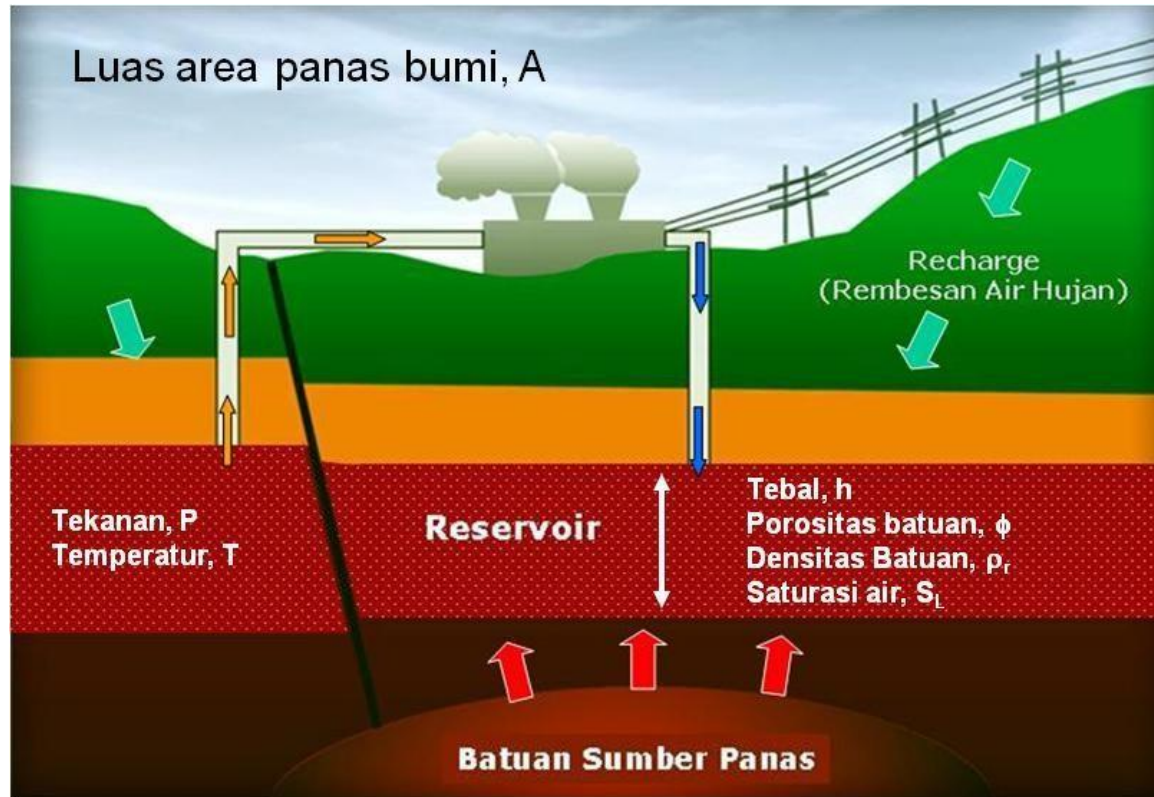
Rapat Daya



Rapat Daya



Metoda Volumetrik (Heat Stored Method)



Panas
yang terkandung
dalam reservoir

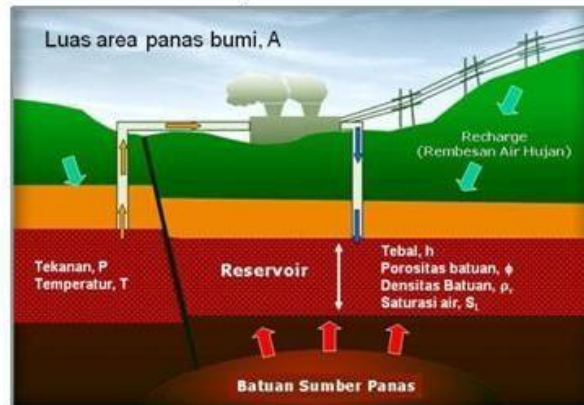
=

Panas
dalam
batuan

+

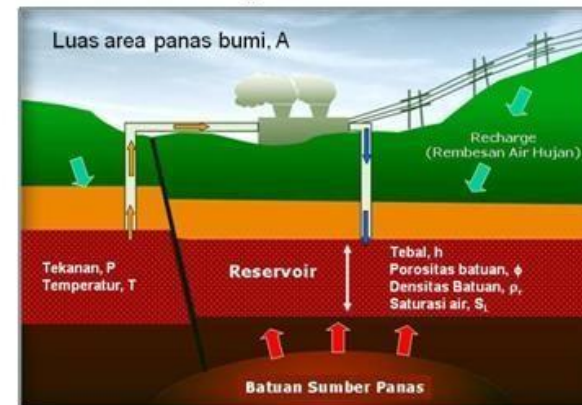
Panas
dalam
fluida

Pada keadaan awal
Temperatur : T_i

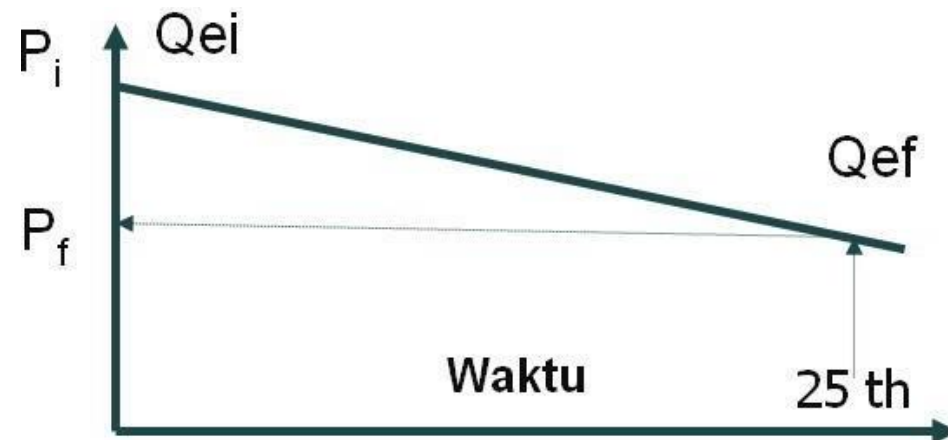


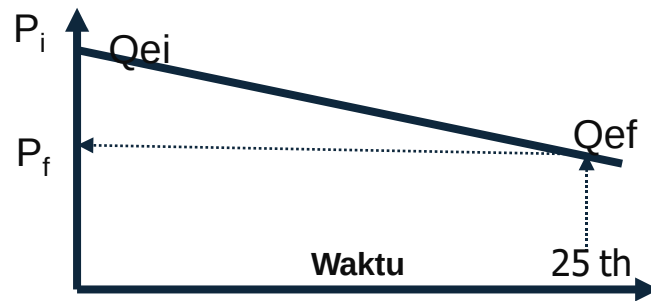
Panas yang tersimpan
dalam reservoir : Q_{ei}

Pada keadaan awal
Temperatur : T_f



Panas yang tersimpan
dalam reservoir : Q_{eif}

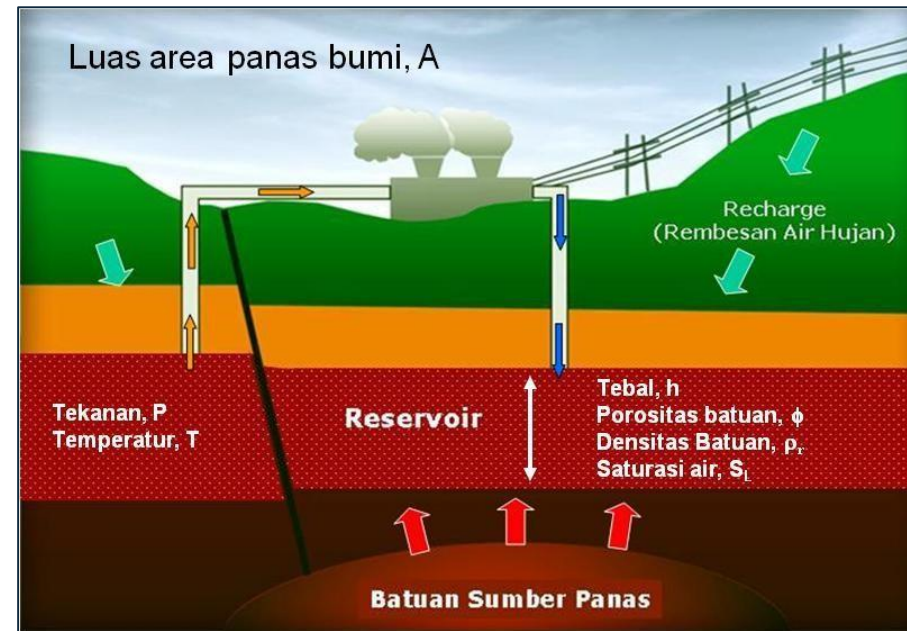




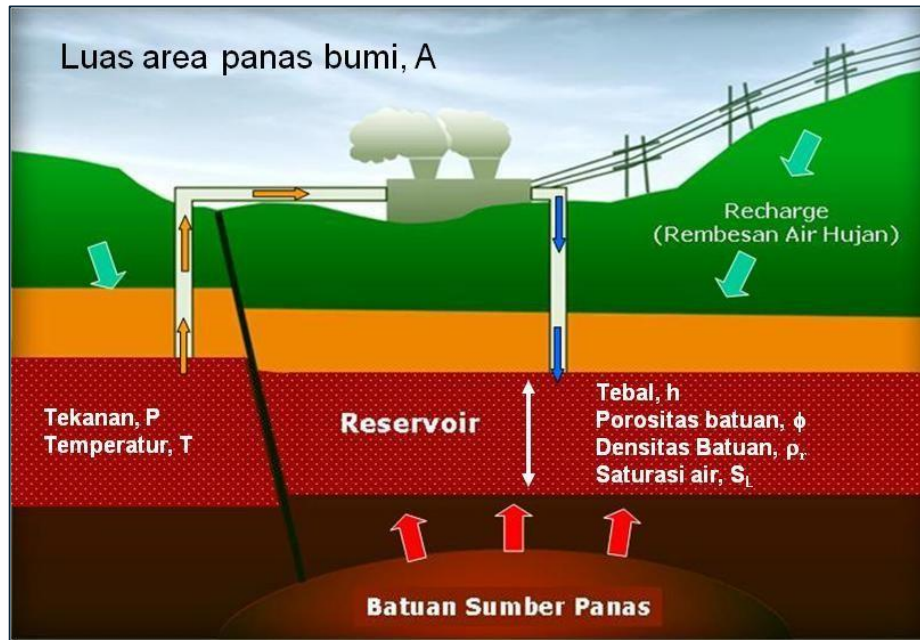
- Panas maksimum yang dapat dimanfaatkan, $Q_{th} = Q_{ei} - Q_{ef}$
- Bila RF = Faktor perolehan
- Panas yang dapat diproduksi, $Q_{de} = RF \times Q_{th}$
- Panas yang dapat dimanfaatkan selama waktu t:

$$Q_{thermal} = \frac{Q_{de}}{t \times 365 \times 24 \times 3600}$$

Potensi Listrik, $Q_e = \text{Faktor Konversi listrik} \times Q_{thermal}$



$$\begin{aligned}
 &\text{Panas yang terkandung dalam reservoir yang dapat dimanfaatkan} \\
 &= \text{Panas dalam Batuan yang dapat dimanfaatkan} + \text{Panas dalam Fluida yang dapat dimanfaatkan} \\
 &= m_r c_r (T_i - T_f) \\
 &= [\text{kg}][\text{kJ/kg}^\circ\text{K}][^\circ\text{K}]
 \end{aligned}$$



Masa matriks batuan

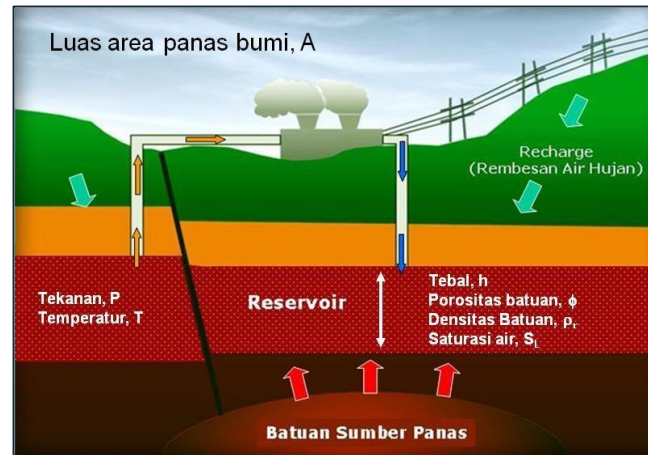
$$\rho_r = m_r / V_r \rightarrow m_r = \rho_r V_r$$

$$m_r = \rho_r (1 - \phi) V$$

Panas dibatuan yang dapat dimanfaatkan = $m_r c_r (T_i - T_f)$

$$= \rho_r (1 - \phi) V c_r (T_i - T_f)$$

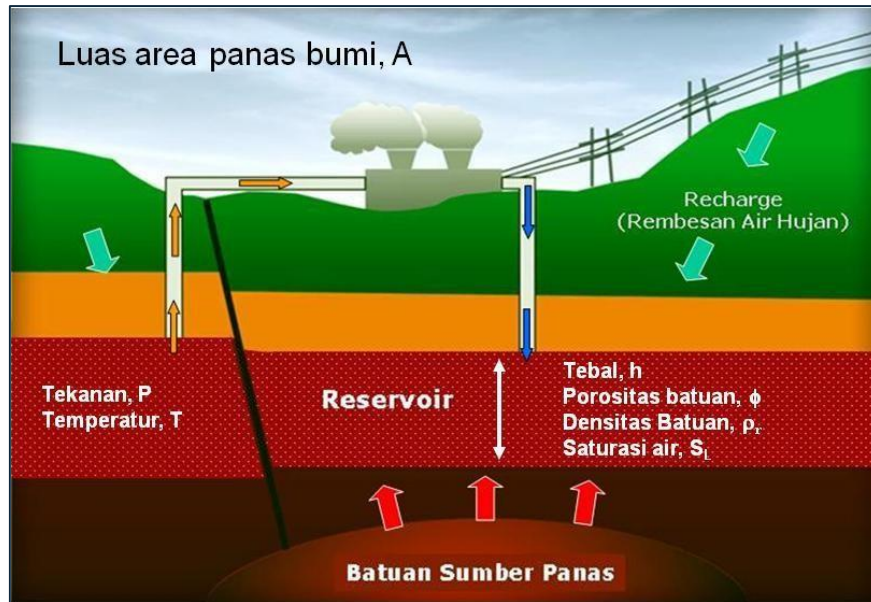
$$V = A h \quad \Rightarrow \quad = A h (1 - \phi) \rho_r c_r (T_i - T_f)$$



Panas yang terkandung dalam reservoir yang dapat dimanfaatkan	=	<u>Panas dalam Batuan yang dapat dimanfaatkan</u>	+	<u>Panas dalam Fluida yang dapat dimanfaatkan</u>
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$$= (m_L u_L + m_V u_V)_i - (m_L u_L + m_V u_V)_f$$

$$[\text{Satuan: kg}][\text{kJ/kg}]$$

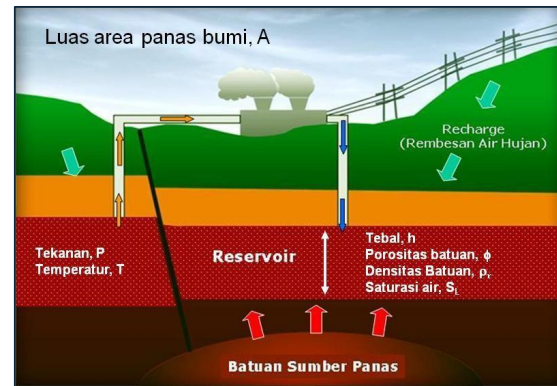


Panas yang tersimpan dalam fluida

$$= A h \phi S_L \rho_L u_L + A h \phi S_v \rho_v u_v$$

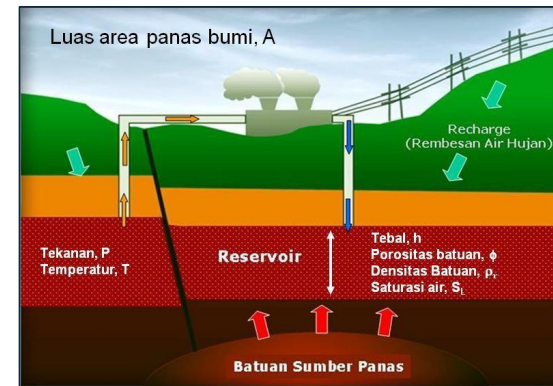
$$= A h \phi (S_L \rho_L u_L + S_v \rho_v u_v)$$

Pada keadaan awal Temperatur : T_i



Panas yang tersimpan dalam fluida pada keadaan awal
 $= Ah \phi (S_L \rho_L u_L + S_v \rho_v u_v)_i$

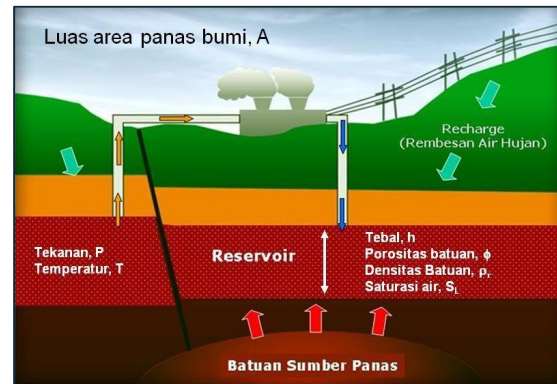
Pada keadaan akhir Temperatur : T_f



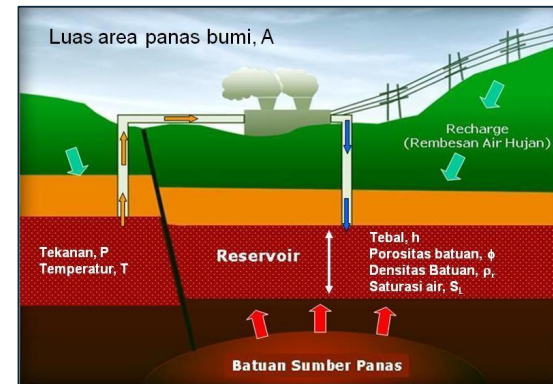
Panas yang tersimpan dalam fluida pada keadaan akhir
 $= Ah \phi (S_L \rho_L u_L + S_v \rho_v u_v)_f$

Panas yang tersimpan dalam fluida yang dapat dimanfaatkan
 $= Ah \phi [(S_L \rho_L u_L + S_v \rho_v u_v)_i - (S_L \rho_L u_L + S_v \rho_v u_v)_f]$

Pada keadaan awal Temperatur : T_i



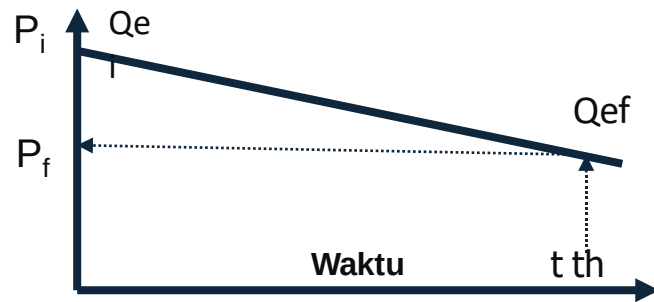
Pada keadaan akhir Temperatur : T_f



Panas yang tersimpan dalam batuan dan fluida yang dapat dimanfaatkan

$$Q_{th} = Ah(1-\phi) \rho_r c_r (T_i - T_f) + Ah \phi [(S_L \rho_L u_L + S_v \rho_v u_v)_i - (S_L \rho_L u_L + S_v \rho_v u_v)_f]$$

$$Q_{th} = Ah\{(1-\phi) \rho_r c_r (T_i - T_f) + \phi [(S_L \rho_L u_L + S_v \rho_v u_v)_i - (S_L \rho_L u_L + S_v \rho_v u_v)_f]\}$$



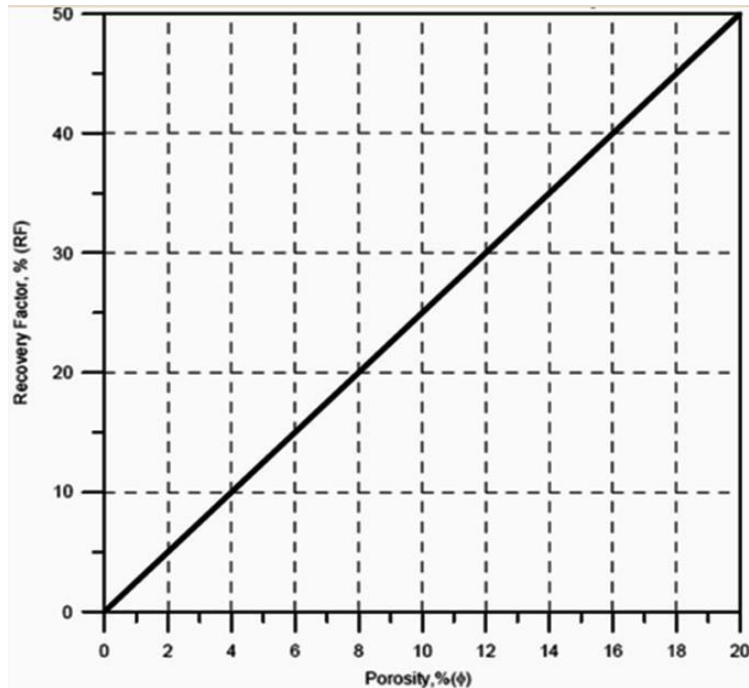
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- Bila RF = Faktor perolehan
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- Panas yang dapat dimanfaatkan selama waktu t tahun:

$$Q_{thermal} = \frac{Q_{de}}{t \times 365 \times 24 \times 3600}$$

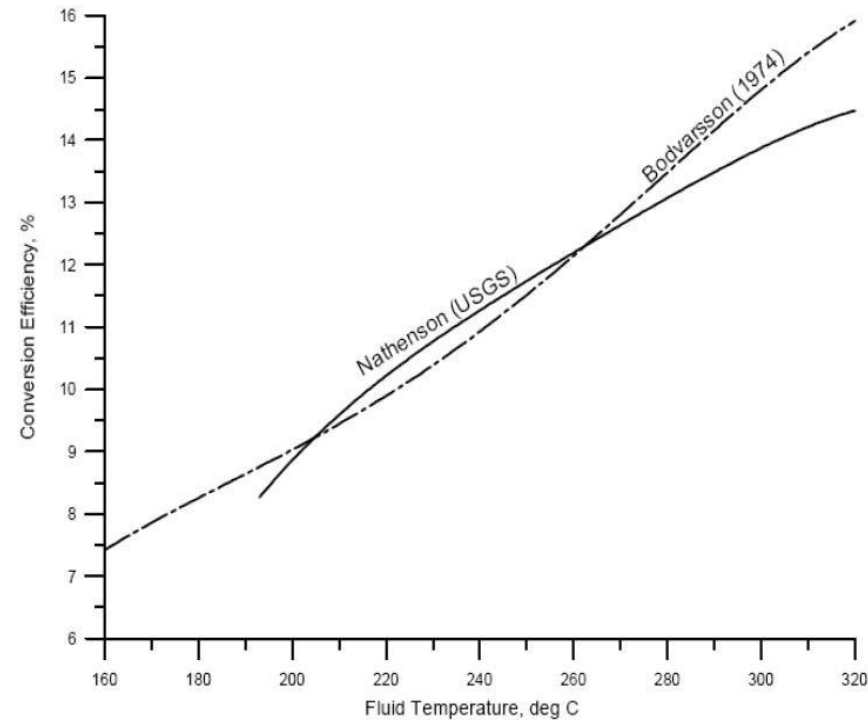
Satuan: kJ/det = kWatt

Potensi Listrik, $Q_e = \text{Faktor Konversi listrik} \times Q_{thermal}$

Recovery Factor (RF) & Faktor Konversi (η)



Faktor perolehan didapatkan dari korelasi dengan porositas berdasarkan grafik After Muffler 1978



Korelasi faktor konversi listrik dengan temperatur fluida menurut Nathenson (1975) dan Bodvarsson (1974)

Recovery Factor (RF)

Table 7.1 Default values of recovery factor recommended by the Australian code

Type of reservoir	Mean value of R_f	Range of R_f
Fracture dominated reservoirs (including EGS)	14%	8–20% with a uniform probability
Sedimentary reservoirs or porous volcanic-hosted reservoirs, of “moderate” porosity (<7% on average)	17.5%	10–25% with a uniform probability
Sedimentary or porous volcanic-hosted reservoir of exceptionally high average porosity (>7%)	2.5 times the porosity	Up to a maximum of 50%

O’Sullivan, M. J., and O’Sullivan, J. P. (2016): Reservoir modeling and simulation for geothermal resource characterization and evaluation, *Geothermal Power Generation: Developments and Innovation*, Elsevier Ltd, 165–199. <https://doi.org/10.1016/B978-0-08-100337-4.00007-3>

SNI 6482: 2018 Parameter dalam estimasi potensi energi panas bumi

Tabel 2 – Nilai parameter pada metode volumetrik

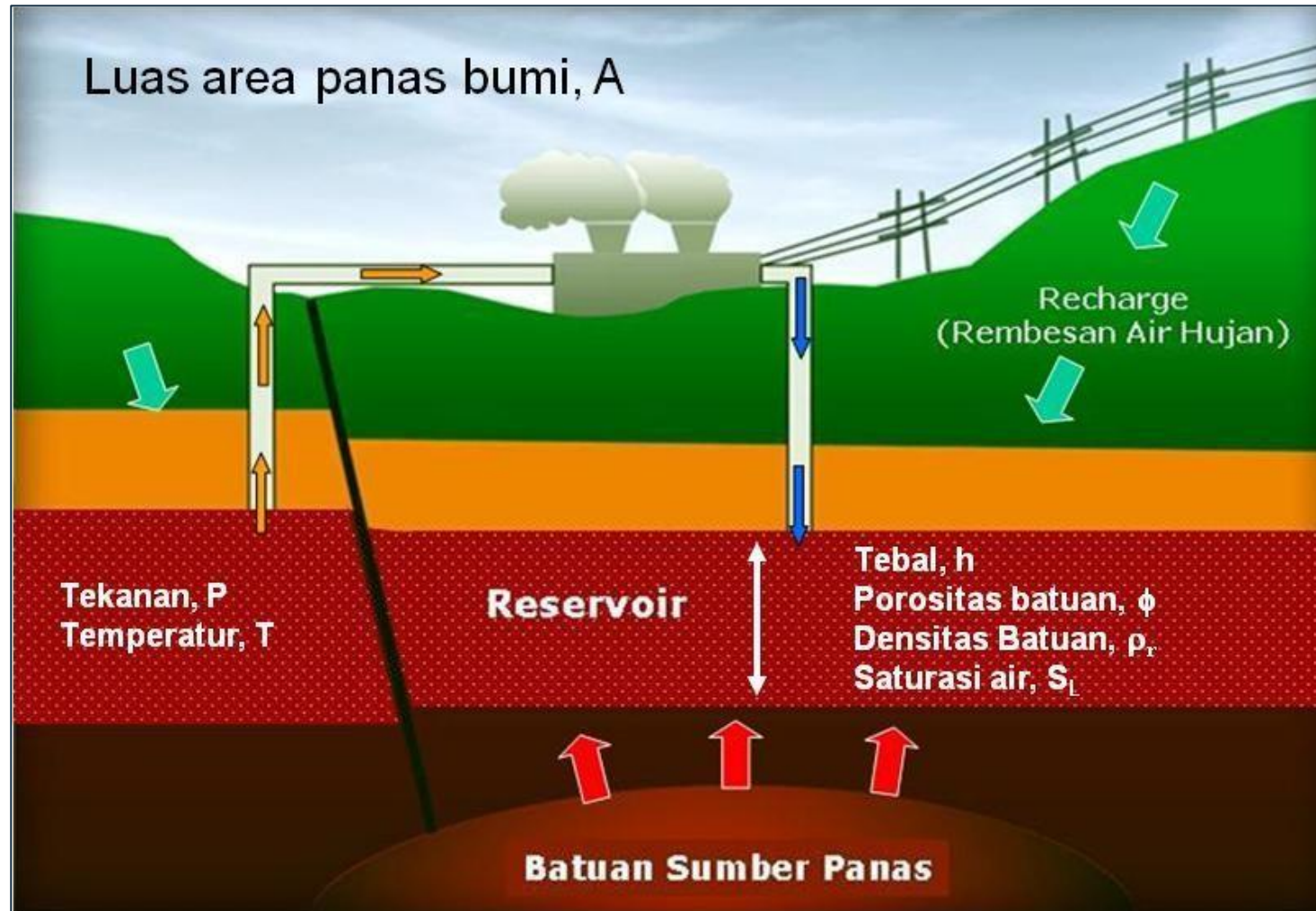
Parameter	Temperatur rendah (<125 °C)	Temperatur sedang (125 - 225 °C)	Temperatur tinggi (>225 °C)
Saturasi air (%)	100	80	70
Porositas batuan reservoir (%)	12	10	7
Kapasitas panas batuan (kJ/kg C)	0,8	0,9	1,0
Densitas batuan (kg/m ³)	$2,65 \times 10^3$	$2,65 \times 10^3$	$2,65 \times 10^3$
Umur proyek (tahun)	30	30	30
Faktor konversi listrik (%)	7	10	10
<i>Recovery factor</i> (%)	20	25	30

Tabel 3 - Nilai parameter pada metode kandungan panas

Parameter	Temperatur rendah (<125 °C)	Temperatur sedang (125 - 225 °C)	Temperatur tinggi (>225 °C)
Umur proyek (tahun)	30	30	30
Faktor konversi listrik (%)	7	10	10
<i>Recovery factor</i> (%)	20	25	30

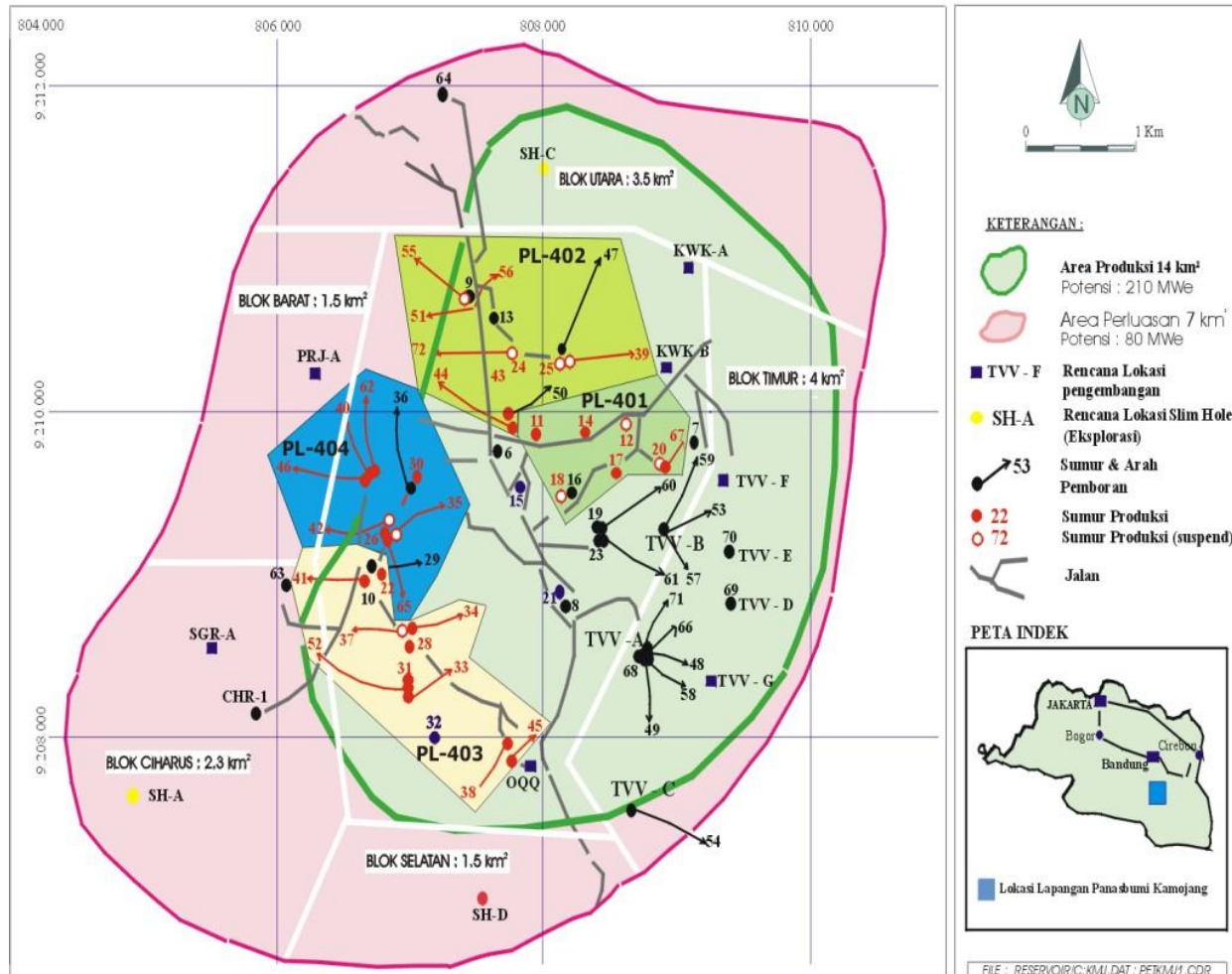
	Awal	Akhir
Area (sq-km)=		
Tebal (km)=		
Porositas (fraksi)=		
Kap. Panas batuan (kJ/kg.deg-C)=		
Densitas batuan (kg/cu-m)=		
Sat. liquid (fraksi)=		
Sat. uap (fraksi)=		
Temp. res (deg-C)=		
Project time (tahun)=		
El. Conv. Factor (%)=		
Rec. Factor (%)=		
H rock (kJ)=		
H air (kJ)=		
H uap (kJ)=		
H total (kJ)=		
Energi tersedia (kJ)=		
Energi yang bisa diambil (kJ)=		
MW_{elektrik} =		

Kelemahan Metoda Heat Stored ?



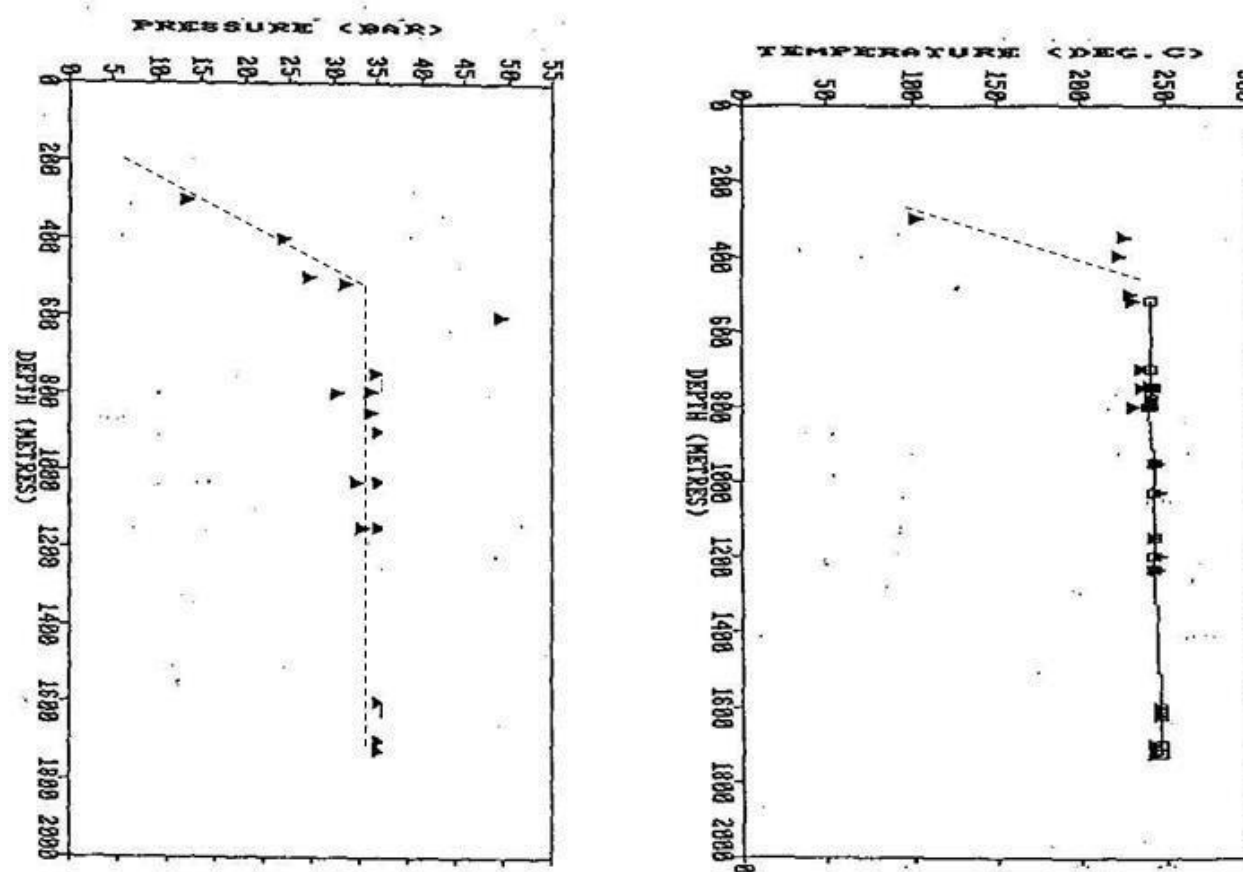
Studi Kasus : Perhitungan Potensi Listrik Kamojang

LAPANGAN PANASBUMI KAMOJANG



- Lapangan panasbumi Kamojang merupakan reservoir dominasi uap. Pelamparan daerah panasbumi serta lokasi sumur sbb:

Dari data tekanan dan temperatur di sejumlah sumur, diperoleh profil tekanan dan temperatur seperti diperlihatkan pada gambar berikut:



Temperatur Reservoir : 235 - 245°C
Tekanan Reservoir : 34 - 35 bar

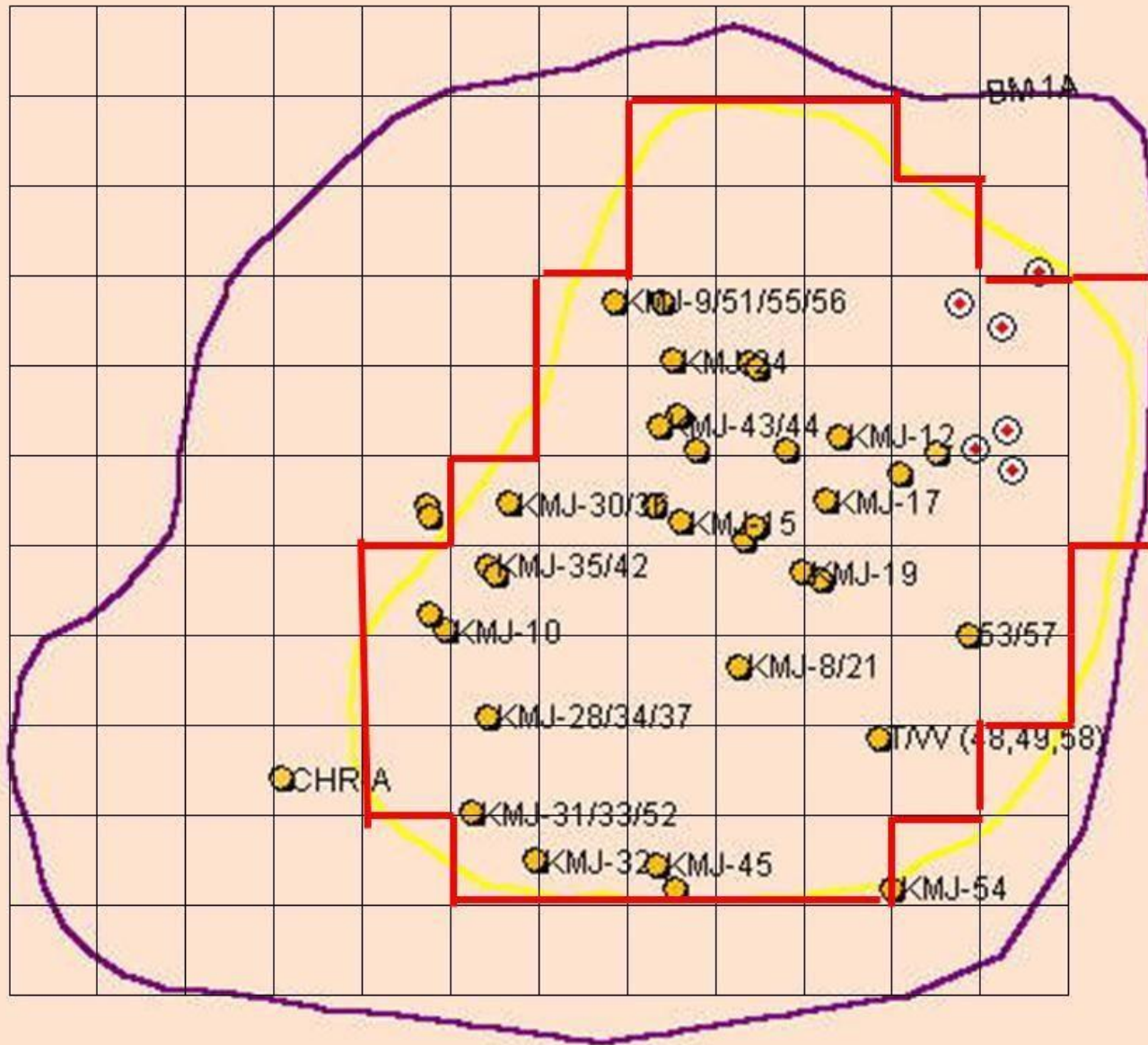
Contoh soal

1. Hitunglah potensi listrik dari lapangan tersebut. Gunakan asumsi berikut:

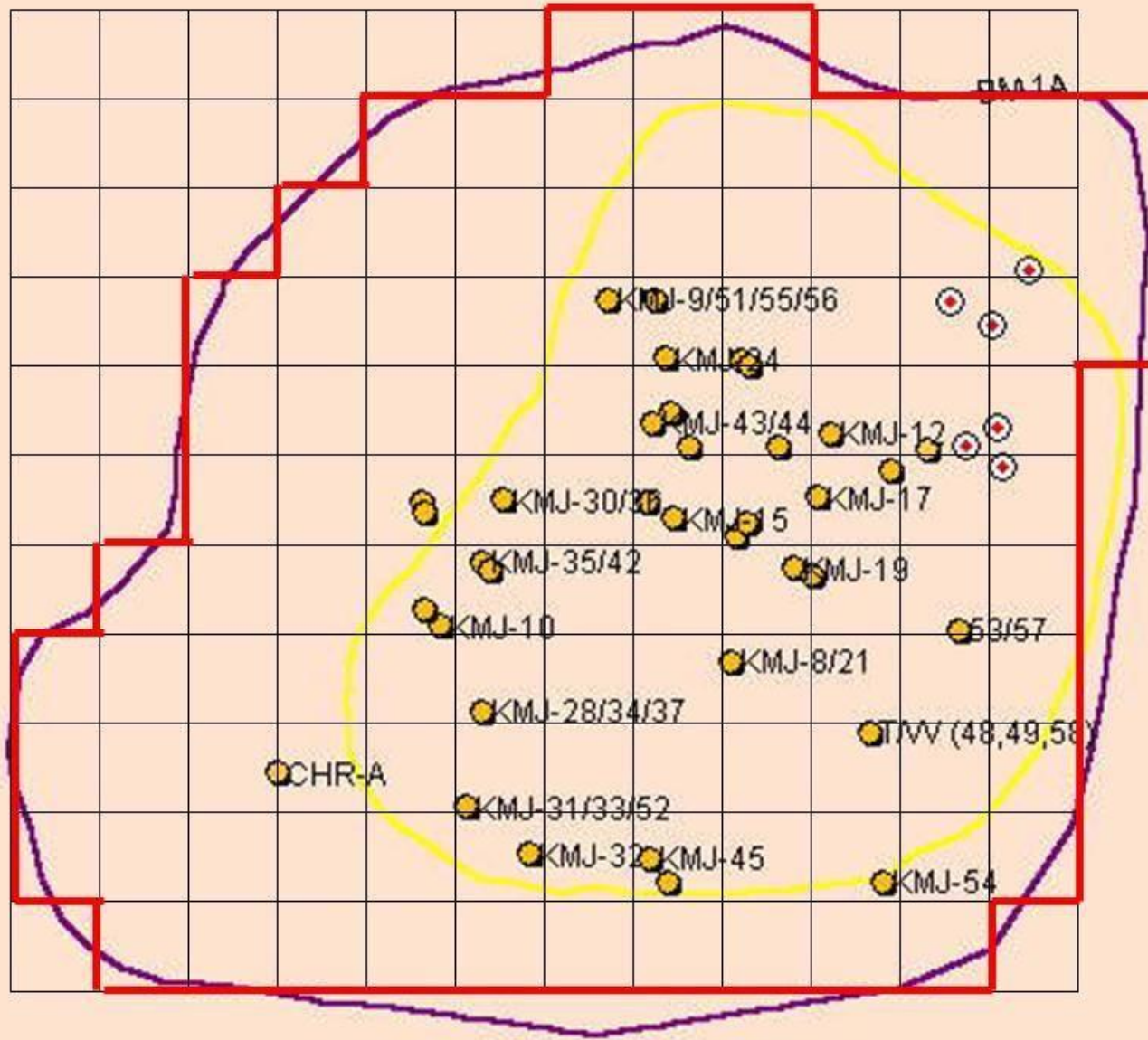
Sifat petrofisik batuan:

- Porositas = 10%
- Densitas = 2500 kg/m³
- Panas spesifik = 1000 J/kg⁰C
- Tebal zona uap = 2000 m
- Faktor Perolehan = 30%
- Faktor Konversi Listrik = 15%

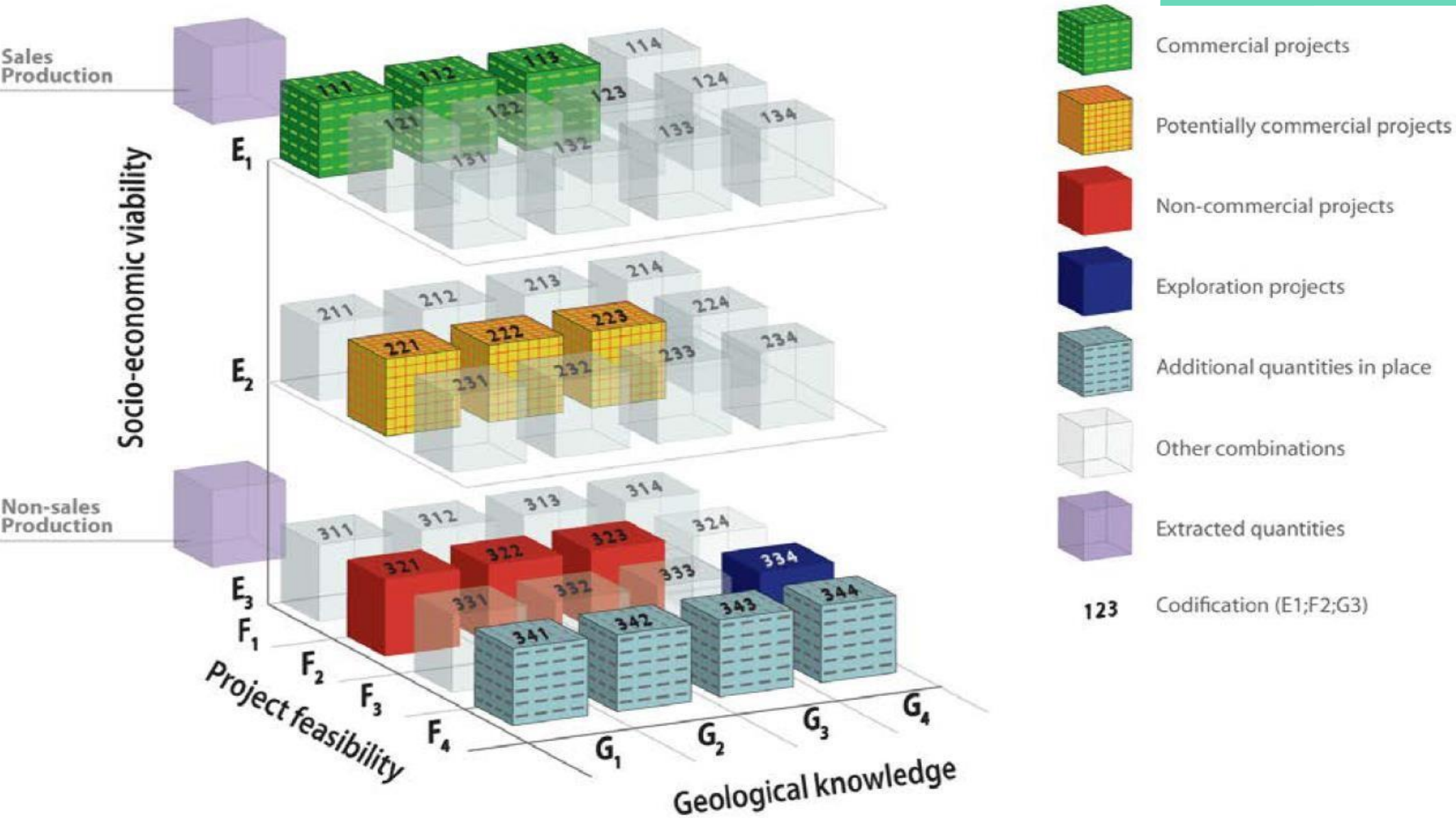
2. Pada saat ini kapasitas listrik terpasang di PLTP Kamojang adalah 235 MW_e. Kapasitas PLTP ini akan ditingkatkan. Menurut saudara, apakah mungkin untuk meningkatkan kapasitasnya menjadi 290 MW_e ? Berikan alasan saudara ! Bila tidak mungkin, berapakah besar kapasitas PLTP maksimum menurut saudara. Berikan dasar pertimbangannya!



$A = 14 \text{ km}^2$



Luas Area = 20 km²



UNFC Classes Defined by Categories and Sub-categories						
Total Commodity Initially in Place	Extracted	Sales Production				
		Non-sales Production				
	Class		Sub-class	Categories		
				E	F	G
	Known Deposit	Commercial Projects	On Production	1	1.1	1, 2, 3
			Approved for Development	1	1.2	1, 2, 3
			Justified for Development	1	1.3	1, 2, 3
		Potentially Commercial Projects	Development Pending	2 ^b	2.1	1, 2, 3
			Development On Hold	2	2.2	1, 2, 3
		Non-Commercial Projects	Development Unclassified	3.2	2.2	1, 2, 3
			Development Not Viable	3.3	2.3	1, 2, 3
		Additional Quantities in Place		3.3	4	1, 2, 3
	Potential Deposit	Exploration Projects	[No sub-classes defined] ^c	3.2	3	4
		Additional Quantities in Place		3.3	4	4

Category	Definition ^b	Supporting Explanation ^c
E1	Extraction and sale has been confirmed to be economically viable. ^d	Extraction and sale is economic on the basis of current market conditions and realistic assumptions of future market conditions. All necessary approvals/contracts have been confirmed or there are reasonable expectations that all such approvals/contracts will be obtained within a reasonable timeframe. Economic viability is not affected by short-term adverse market conditions provided that longer-term forecasts remain positive.
E2	Extraction and sale is expected to become economically viable in the foreseeable future. ^d	Extraction and sale has not yet been confirmed to be economic but, on the basis of realistic assumptions of future market conditions, there are reasonable prospects for economic extraction and sale in the foreseeable future.
E3	Extraction and sale is not expected to become economically viable in the foreseeable future or evaluation is at too early a stage to determine economic viability. ^d	On the basis of realistic assumptions of future market conditions, it is currently considered that there are not reasonable prospects for economic extraction and sale in the foreseeable future; or, economic viability of extraction cannot yet be determined due to insufficient information (e.g. during the exploration phase). Also included are quantities that are forecast to be extracted, but which will not be available for sale.

Category	Definition	Supporting Explanation
F1	Feasibility of extraction by a defined development project or mining operation has been confirmed.	Extraction is currently taking place; or, implementation of the development project or mining operation is underway; or, sufficiently detailed studies have been completed to demonstrate the feasibility of extraction by implementing a defined development project or mining operation.
F2	Feasibility of extraction by a defined development project or mining operation is subject to further evaluation.	Preliminary studies demonstrate the existence of a deposit in such form, quality and quantity that the feasibility of extraction by a defined (at least in broad terms) development project or mining operation can be evaluated. Further data acquisition and/or studies may be required to confirm the feasibility of extraction.
F3	Feasibility of extraction by a defined development project or mining operation cannot be evaluated due to limited technical data.	Very preliminary studies (e.g. during the exploration phase), which may be based on a defined (at least in conceptual terms) development project or mining operation, indicate the need for further data acquisition in order to confirm the existence of a deposit in such form, quality and quantity that the feasibility of extraction can be evaluated.
F4	No development project or mining operation has been identified.	In situ (in-place) quantities that will not be extracted by any currently defined development project or mining operation.

G1	Quantities associated with a known deposit that can be estimated with a high level of confidence.	For in situ (in-place) quantities, and for recoverable estimates of fossil energy and mineral resources that are extracted as solids, quantities are typically categorised discretely, where each discrete estimate reflects the level of geological knowledge and confidence associated with a specific part of the deposit. The estimates are categorised as G1, G2 and/or G3 as appropriate. For recoverable estimates of fossil energy and mineral resources that are extracted as fluids, their mobile nature generally precludes assigning recoverable quantities to discrete parts of an accumulation. Recoverable quantities should be evaluated on the basis of the impact of the development scheme on the accumulation as a whole and are usually categorised on the basis of three scenarios or outcomes that are equivalent to G1, G1+G2 and G1+G2+G3.
G2	Quantities associated with a known deposit that can be estimated with a moderate level of confidence.	
G3	Quantities associated with a known deposit that can be estimated with a low level of confidence.	

Category	Definition	Supporting Explanation
G4	Estimated quantities associated with a potential deposit, based primarily on indirect evidence.	Quantities that are estimated during the exploration phase are subject to a substantial range of uncertainty as well as a major risk that no development project or mining operation may subsequently be implemented to extract the estimated quantities. Where a single estimate is provided, it should be the expected outcome but, where possible, a full range of uncertainty in the size of the potential deposit should be documented (e.g. in the form of a probability distribution). In addition, it is recommended that the chance (probability) that the potential deposit will become a deposit of any commercial significance is also documented.

Resource Assessment: Volumetric Method

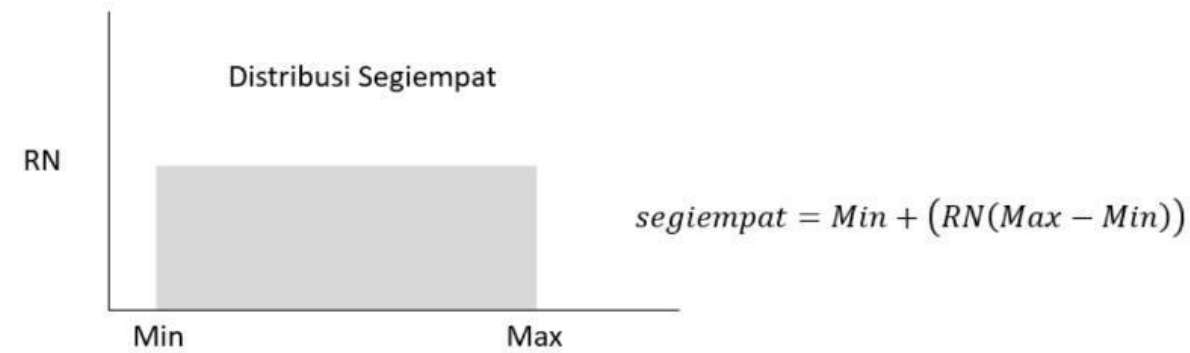
Heat Stored: Monte Carlo Simulation

Input Parameters

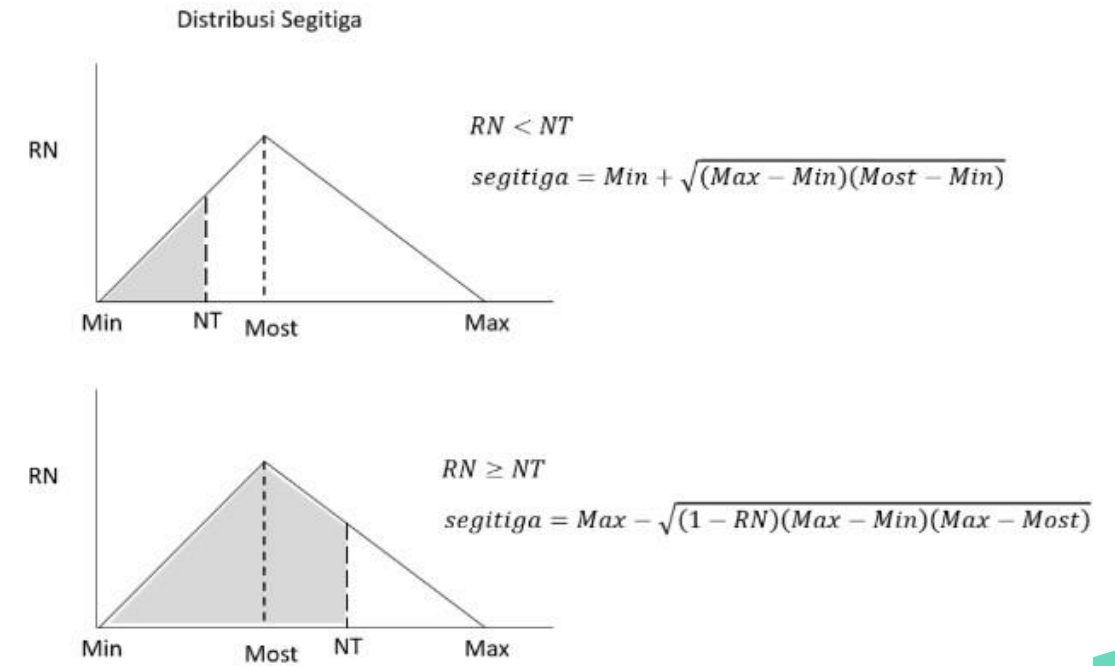
- Area [A] (km^2)
- Reservoir Thickness [h] (m)
- Porosity [ϕ] (fraction)
- Rock Density [ρ_{rock}] (kg/m^3)
- Heat Rock Capacity [Cr] ($\text{kJ}/\text{kg}^\circ\text{C}$)
- Initial Temperature [T_i] ($^\circ\text{C}$)
- Final Temperature [T_f] ($^\circ\text{C}$)
- Initial Water Saturation [S_{w_i}] (fraction)
- Final Water Saturation [S_{w_f}] (fraction)
- Recovery Factor [RF] (fraction)
- Electricity Conversion Factor (fraction)
- Lifetime (years)

Frequency Distribution

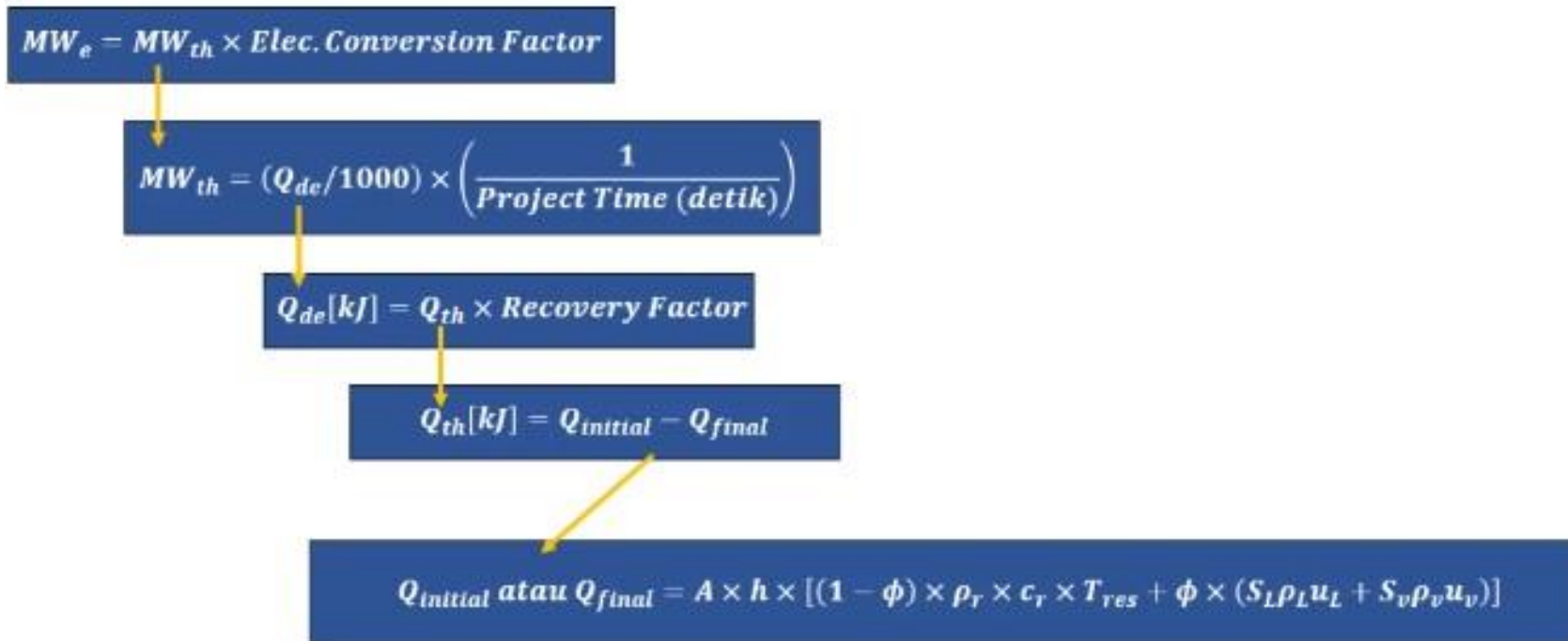
- Distribusi Segi Empat (*Uniform Distribution*)



- Distribusi Segitiga (*Triangular Distribution*)



Calculation Procedure



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